18 Mark for Review

One gallon of paint will cover 310 square feet of a surface. A room has a total wall area of w square feet. Which equation represents the total amount of paint P, in gallons, needed to paint the walls of the room twice?

(A) $P = \frac{w}{155}$

(B) $P = 620w$

(C) $P = \frac{w}{310}$

(D) $P = 310w$

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16 Mark for Review

A school is ordering tablet computers for its students. The tablets cost $164 each, and 6% tax is added to the total cost of the order. If the school can spend no more than 44,800 dollars on the tablets, including tax, what is the maximum number of tablets that the school can order?

Answer Preview:

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$$\frac{x}{7} + \frac{y}{9} = \frac{41}{63}$$

An engineer connects resistors in series, where the resistors in the series have a total resistance of $\frac{41}{63}$ ohms. In this series, there are x resistors of type A, which each have a resistance of a ohms, and y resistors of type B, which each have a resistance of b ohms. The given equation represents this situation. According to this equation, what is the positive difference between the value of a and the value of b?

A. 41

B. 2

C. $\frac{41}{63}$

D. $\frac{2}{63}$

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A beekeeper's initial observation of the population of a certain bee colony was $1{,}800$ bees. The beekeeper set a goal to increase the population to $3{,}300$ bees. The beekeeper uses a model that predicts the population of this bee colony begins at $1{,}800$ and increases by 120 bees per week in the first two weeks after the initial observation, and then increases by 180 bees per week until the beekeeper's goal is reached. According to this model, at the end of week w after the initial observation, where $w > 2$, which of the following functions gives the predicted number of bees still needed to reach the beekeeper's goal?

A. $p(w) = 3{,}300 - 180w$

B. $p(w) = 3{,}180 + 180w$

C. $p(w) = 1{,}620 - 180w$

D. $p(w) = -120 + 180w$

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2 Mark for Review

Hannah and Wyatt are saving money to purchase a car. Hannah saves $\frac{1}{5}$ of her salary each month, and Wyatt saves $\frac{2}{7}$ of his salary each month. Together, they save a total of $\$3,270$ from their monthly salaries each month. If h and w represent Hannah's and Wyatt's monthly salaries, in dollars, respectively, which equation shows the relationship between h and w?

(A) $h + w = 3,270$

(B) $h + 2w = 3,270$

(C) $10h + 7w = 114,450$

(D) $7h + 10w = 114,450$

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10 Mark for Review

$$5(kx - n) = -\frac{65}{14}x - \frac{85}{18}$$

In the given equation, k and n are constants and $n > 1$. The equation has no solution. What is the value of k?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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17 Mark for Review

One gallon of sealant costs $13 and will cover 300 square feet of a surface. A deck has a total surface area of d square feet. Which equation represents the cost c, in dollars, of the sealant needed to cover the deck twice?

(A) $c = \frac{300d}{13}$

(B) $c = \frac{600d}{13}$

(C) $c = 13\left(\frac{d}{150}\right)$

(D) $c = 13\left(\frac{d}{300}\right)$

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393 In the *xy*-plane, which of the following does NOT contain any points (x, y) that are solutions to the inequality $-5x < 70y - 75$?

A. The region where $x < 0$ and $y > 0$

B. The region where $x < 0$ and $y < 0$

C. The region where $x > 0$ and $y > 0$

D. The region where $x > 0$ and $y < 0$

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